Max Otto Jones — Curriculum Vitae

PERSONAL

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INTERESTS

Solving the many problems that occur in a biological system requires an understanding of how the system works. To understand a system, we must understand, at multiple levels of analysis, the principles by which it operates. How can we design careful experiments and tools to unveil these principles? A good attempt at answering this question does not rely on a singular approach, but creates meaningful connections between them.

EDUCATION

Carleton College, Northfield, MN
Bachelor of Arts in Cognitive Science (expected 2028)
Neuroscience minor. Mathematics minor. 3.89 GPA.

RESEARCH EXPERIENCE

Research Assistant, PI: **Dr. Joel A. Tripp**
Neuroscience Program, Carleton College, Northfield, MN
September 2025 - Present

- Investigating the neural circuitry of the vocal-motor system in Alston's singing mice to determine how neurobiological substrates mediate socially relevant behavior.
- Conducting neuropeptide anatomy experiments to map modulatory influences on vocal-motor circuits, advancing understanding of the neurochemical basis of social behaviors.
- Analyzing and visualizing spectrogram vocalization data

Research Assistant, PI: **Dr. David J. Herzfeld**, Co-Mentor: **Dr. Aaron Suminski**
University of Wisconsin School of Medicine & Public Health, Madison, WI
November 2025 - Present (May-Aug 2026 funded by *National Science Foundation* and *College of Engineering*)

- Designing and conducting novel behavioral experiments in rhesus monkeys to dissect the cerebellar basis of motor memories using extracellular electrophysiology.
- Designed a converter for visualization of Full Binary Pursuit spike-sorting outputs in Phy, resolving a long-standing integration barrier in the lab's electrophysiology pipeline.
- Revealed a time-delay property of action potentials that allows for exhaustive detection in extracellular recordings, reducing the sensitivity-specificity tradeoff that limits reproducibility in systems neuroscience.

PUBLICATIONS

Jones M.O.†, and **Herzfeld, D.J**. Exploiting deterministic dynamics of action potentials for precise and exhaustive spike sorting. In Preparation.

Szycher A.D.†, **Jones, M.O.†**, Janik J.R., and **Tripp, J.A**. Distribution of vasopressin and oxytocin neurons in Alston's singing mouse. In Preparation.

Bond, R.†, Brown, L., **Jones, M.O.**, and **Tripp, J.A**. Ghrelin suppresses courtship behavior in male mice. In Preparation.

CONFERENCE PRESENTATIONS

Jones, M.O., and **Herzfeld, D. J.** Exploiting deterministic dynamics of action potentials for precise and exhaustive spike sorting. *Soc. Neurosci*, Washington D.C.. 2026. Presented.

Bond, R.†, Brown, L., **Jones, M.O.**, and **Joel A. Tripp**. Hunger Hormone Ghrelin Suppresses Courtship Behavior in Male Mice. *Soc. Behav. Neuroendocrinology*, Niagara Falls, NY. 2026. Presented.

Szycher A.D., **Jones, M.O.**, Janik J.R., and **Joel A. Tripp**. Characterizing the distribution of vasopressin and oxytocin neurons and their relation to vocal-motor circuitry in Alston's singing mouse. *Midbrains Conf.*, Sioux Falls, SD. 2025. Presented.

Szycher A.D., **Jones, M.O.**, Janik J.R., and **Joel A. Tripp**. Characterizing the distribution of vasopressin and oxytocin neurons and their relation to vocal-motor circuitry in Alston's singing mouse. *Soc. Behav. Neuroendocrinology*, Online. 2025. Attended.

RELEVANT COURSEWORK

Neuro/Bio: Foundations in Neuroscience (w Lab), Neurons Circuits & Behavior (w Lab)
Seminar: Topics in Neuroscience, Genes Evolution & Development (w Lab)
Math/CS: Linear Algebra, Intro to Computer Science, Data Structures, Psychological Statistics

TECHNICAL SKILLS

Programming: Python, Kotlin, R, Julia
Experimental Techniques: Confocal Laser Microscopy, Multi-Contact NHP Neurophysiology, Spike Sorting, Behavioral Data Collection, Rodent Brain Cryosectioning, Fluorescent Immunohistochemistry
Tools: Spikeinterface, Matplotlib, UnitRefine, Bombcell, NumPy, SciPy, Phy, Kilosort, MountainSort, SpyKing Circus, Full Binary Pursuit, DeepSqueak

HONORS & AWARDS

2026, NSF SURE-REU Fellow, UW-Madison College of Engineering.
2026, NCAA DIII All-Conference Honors, Track & Field.
2025, Towsley Conference Travel Award, Carleton College.
2024, Academic All-State Honors, Football.
2024, DI All-Conference Honors, Baseball.
2023, DI All-State Honors, School Record, Swimming.

TEACHING EXPERIENCE

Sophomore Spring, *Hormones, Brain, and Behavior*
Carleton College, Teaching Assistant & Grader

LANGUAGES

English (native), German (proficient)